Binary Market Data Format Specification

Complete Data Structure Layout for FEED_DETAILS Stream Engine

File Type: Binary Capture (.bin)
Byte Order: Little-Endian
Payload Struct: FEED_DETAILS
Depth Level: FEED_LEVEL_DEPTH = 5
Struct Size: 216 Bytes (Unpacked)
Version: 2.0 (Clean Build)

1. Technical Summary

This specification defines the byte-level protocol for high-frequency Level-2 order book persistence. Market depth updates and execution events are serialized sequentially, pairing core instrument identification metadata with precise distribution latency timestamps.

> **Structural Features**
>
> The FEED_DETAILS architecture minimizes runtime parsing overhead by combining symbol metadata, order depth levels (top 5 bids and asks), and end-to-end timing metrics (Source, Distribution Receive, and Distribution Send timestamps) into a single contiguous structure.

2. Stream Framing & Framing Envelope

The binary capture stream uses length-prefixed framing blocks. Each record frame starts with an explicit 8-byte signed integer detailing the length of the binary payload that directly follows.

BINARY STREAM ENVELOPE STRUCT

Header: nRead 8 Bytes (int64_t)	Payload Block (Length = nRead Bytes) [Capture Timestamp (8B)] + [FEED_DETAILS Data Structure (208B)]

Field Name	Data Type	Size (Bytes)	Description
`nRead`	`int64_t`	8	Length identifier specifying total bytes in the succeeding payload frame.
`RcTime`	`uint64_t`	8	Capture timestamp recorded at stream intake.
`FEED_DETAILS`	`struct`	208	Full market depth update payload.

3. Binary Field Layout & Alignment Mapping

> **Memory Padding Notice**
>
> The structure utilizes compiler default packing (unpacked). On 64-bit architectures, a 4-byte padding offset is automatically inserted following the bitfield members to enforce 8-byte natural alignment for subsequent 64-bit timestamp fields.

Offset	Field Name	Data Type	Bytes	Description
0x000	`SymbolCode`	`char[50]`	50	Null-terminated ticker symbol string identifier
0x032	`Index`	`int32_t`	4	Internal instrument index key
0x036	`ExpiryDate`	`int32_t`	4	Contract expiration date format (YYYYMMDD)
0x03A	`BuyPrice[5]`	`int32_t[5]`	20	Bid prices for top 5 market depth levels
0x04E	`BuyQty[5]`	`uint32_t[5]`	20	Bid quantities for top 5 market depth levels
0x062	`SellPrice[5]`	`int32_t[5]`	20	Ask prices for top 5 market depth levels
0x076	`SellQty[5]`	`uint32_t[5]`	20	Ask quantities for top 5 market depth levels
0x08A	`NoOfBuyOrds[5]`	`int32_t[5]`	20	Buy order count across top 5 depth levels
0x09E	`NoOfSellOrds[5]`	`int32_t[5]`	20	Sell order count across top 5 depth levels
0x0B2	`LastTradedPrice`	`int32_t`	4	Execution price of the most recent trade
0x0B6	`Bitfield Area`	Bitfields	2	`LastTradedQty:12`, `FeedEventAggressor:2`
0x0B8	*Structure Padding*	`uint8_t[4]`	4	Automatic alignment padding added by compiler
0x0BC	`SrcTimestamp`	`uint64_t`	8	Exchange packet emission timestamp
0x0C4	`DS_rcv_timestamp`	`uint64_t`	8	Distribution server intake timestamp
0x0CC	`DS_snd_timestamp`	`uint64_t`	8	Distribution server dispatch timestamp